

CSIS Report

Team: 64

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1. RTLs of each instruction in the description

AND

D0T0: $AR \leftarrow PC$

D0T1: $IR \leftarrow M[AR]$, $PC \leftarrow PC+1$

D0T2: $D0...D7 \leftarrow \text{Decode } IR(12-14)$, $AR \leftarrow IR(0-11)$

D0T3: $DR \leftarrow M[AR]$

D0T4: $AC \leftarrow AC \wedge DR$, $SC \leftarrow 0$

ADD

D1T0: $AR \leftarrow PC$

D1T1: $IR \leftarrow M[AR]$, $PC \leftarrow PC+1$

D1T2: $D0...D7 \leftarrow \text{Decode } IR(12-14)$, $AR \leftarrow IR(0-11)$

D1T3: $DR \leftarrow M[AR]$

D1T4: $AC \leftarrow AC + DR$, $SC \leftarrow 0$

LDA

D2T0: $AR \leftarrow PC$

D2T1: $IR \leftarrow M[AR]$, $PC \leftarrow PC+1$

D2T2: $D0...D7 \leftarrow \text{Decode } IR(12-14)$, $AR \leftarrow IR(0-11)$

D2T3: $DR \leftarrow M[AR]$

D2T4: $AC \leftarrow DR$, $SC \leftarrow 0$

STA

D3T0: AR $\leftarrow$ PC

D3T1: IR $\leftarrow$ M[AR] , PC $\leftarrow$ PC+1

D3T2: D0...D7 $\leftarrow$ *Decode* IR(12-14) , AR $\leftarrow$ IR(0-11)

D3T3: M[AR] $\leftarrow$ AC , SC $\leftarrow$ 0

BUN

D4T0: AR $\leftarrow$ PC

D4T1: IR $\leftarrow$ M[AR] , PC $\leftarrow$ PC+1

D4T2: D0...D7 $\leftarrow$ *Decode* IR(12-14) , AR $\leftarrow$ IR(0-11)

D4T3: PC $\leftarrow$ AR , SC $\leftarrow$ 0

MUL

D5T0: AR $\leftarrow$ PC

D5T1: IR $\leftarrow$ M[AR] , PC $\leftarrow$ PC+1

D5T2: D0...D7 $\leftarrow$ *Decode* IR(12-14) , AR $\leftarrow$ IR(0-11)

D5T3: DR $\leftarrow$ M[AR]

D5T4: AC $\leftarrow$ AC*DR , SC $\leftarrow$ 0

ISZ

D6T0: AR $\leftarrow$ PC

D6T1: IR $\leftarrow$ M[AR] , PC $\leftarrow$ PC+1

D6T2: D0...D7 $\leftarrow$ *Decode* IR(12-14) , AR $\leftarrow$ IR(0-11)

D6T3: DR $\leftarrow$ M[AR]

D6T4: DR $\leftarrow$ DR+1

D6T5: M[AR] $\leftarrow$ DR , *If* (DR=0) *then* (PC $\leftarrow$ PC+1) , SC $\leftarrow$ 0

2. Logical Expressions for LD, INC, and CLR of each Register

$$\mathbf{AR_LD} = T_0 + T_2$$

$$\mathbf{PC_LD} = D_4T_3$$

$$\mathbf{PC_INC} = T_1 + ((DR(0) + DR(1) + \dots + DR(15))')(D_6T_5)$$

$$\begin{aligned}\mathbf{AC_LD} &= D_0T_4 + D_1T_4 + D_2T_4 + D_5T_4 \\ &= T_4(D_0 + D_1 + D_2 + D_5)\end{aligned}$$

$$\begin{aligned}\mathbf{DR_LD} &= D_0T_3 + D_1T_3 + D_2T_3 + D_5T_3 + D_6T_3 \\ &= T_3(D_0 + D_1 + D_2 + D_5 + D_6)\end{aligned}$$

$$\mathbf{DR_INC} = D_6T_4$$

$$\mathbf{IR_LD} = T_1$$

$$\begin{aligned}\mathbf{SC_CLR} &= D_0T_4 + D_1T_4 + D_2T_4 + D_3T_3 + D_4T_3 + D_5T_4 + D_6T_5 \\ &= T_4(D_0 + D_1 + D_2 + D_5) + T_3(D_3 + D_4) + D_6T_5\end{aligned}$$

3.Logical Expressions for Memory Read and Write

$$\begin{aligned}\text{Memory Read} &= T1 + D0T3 + D1T3 + D2T3 + D5T3 + D6T3 \\ &= T1 + T3(D0 + D1 + D2 + D5 + D6)\end{aligned}$$

$$\text{Memory Write} = D6T5 + D3T3$$

4. Logical Expressions for the Bus Encoder

Nothing	000	
AR	001	D4T3
PC	010	T0
AC	011	D3T3
DR	100	D6T5
TR	101	
Memory	110	T1 + T3(D0 + D1 + D2 + D5 + D6)
IR	111	T2

5. Logical Expressions for the ALU Encoder

Nothing	0000	
Transfer DR	0001	D2T4
Transfer Bus	0010	
DR + AC	0011	D1T4
AC + Bus	0100	
DR ANDed with AC	0101	D0T4
Complement AC	0110	
Complement Bus	0111	
Multiply DR and AC	1000	D5T4
Multiply AC and bus	1001	